



Name _____

Loops and Randomness

Today's big idea

If you're about to do the same thing twenty times — **don't**. Tell the computer to repeat it.

repeat 4 **CONTROL**

clap

clap

stomp

That's 12 actions written in 4 lines. Change the 4 to a 100 and you've just done 300 actions by typing two characters.

And when you want the computer to surprise you, use **pick random**.

Words to know

Word	What it means
loop	Code that repeats
repeat n	Do this exactly n times
forever	Do this until the game stops. Nothing can go after it
random	A number the computer picks that you can't predict

Golden rule: every **forever** loop needs a **wait** inside it somewhere. Without one, Scratch freezes and you lose your work.

Class activity A – Dancing Sprite

when green flag clicked **EVENTS**

forever	CONTROL
next costume	LOOKS
wait 0.2 seconds	CONTROL
move 5 steps	MOTION

Change the `0.2` until the walk looks right. Add `if on edge, bounce` so it doesn't escape.

Class activity B – Rain

when green flag clicked	EVENTS
forever	CONTROL
go to x: (pick random -240 to 240) y: 180	MOTION
repeat 20	CONTROL
change y by -18	MOTION
wait 0.02 seconds	CONTROL

A raindrop that appears somewhere random along the top and falls. Forever.

Now change the numbers. Fewer repeats + bigger step = faster. Try it.

Your project – things fall from the sky!

Today your project starts looking like a real game.

- ☐ Add a falling object sprite that fits your theme
- ☐ Make it the right size
- ☐ Give it the falling script (same shape as Rain)
- ☐ **Tune the speed** so it's catchable but not boring
- ☐ **Save it**

It doesn't notice your player yet — you can "catch" it and nothing happens. That's next week.

Homework

1. A second thing to catch. Add another falling sprite that looks different. (*Shortcut: right-click your first one and duplicate it — the code comes with it!*)

2. Unpredictable speed. Use `pick random` so each drop falls at a different speed. (Hint: the `wait` can take a random number.)

Bonus challenges

1. Draw shapes with `repeat` + `turn`. A square is `repeat 4 [move 100, turn 90]`. What makes a triangle? A hexagon? What does `repeat 36 [move 10, turn 10]` draw?
 2. Make your falling object spin as it falls.
 3. Make a starfield — five objects falling at five different speeds.
 4. Use `repeat until` instead of `forever` so it stops after ten drops. (Hard! You'll need a way to count, which we learn in two weeks.)
-

If something breaks

- **Flickers at the top and nothing else?** No `wait` inside the falling loop.
 - **Falls once then stops?** The whole thing needs to be inside a `forever`.
 - **Falls upward?** `change y by 18` should be `change y by -18`. Up is positive.
 - **Scratch froze?** A `forever` with no `wait` in it. Reload and add one.
-

Stuck? Put your hand up — that is what we are here for. And always save before you close the lid.